Mathematical Proofs #1

Interesting selections from students

A simple document handout for courses requiring to write proofs such as Mathematics, Engineering and Statistics. By High school students.

"A good talk contains no proofs; a great talk contains no theorems" - Victor Klee

This document was created for the educational purposes of everyone, regardless of age. Human beings should be able to conquer the barriers of reasoning and understanding, the very elements that separate us, as *humans*, into something extraordinary and precious.

Humans are extraordinary, beings of immeasurable worth and brilliance.
And yet, in a world where billions of animals slave their lives and finally die because of that very distinction, we must remember: human life, in every essence and every scale, is a miracle, fragile, rare, and profoundly precious.

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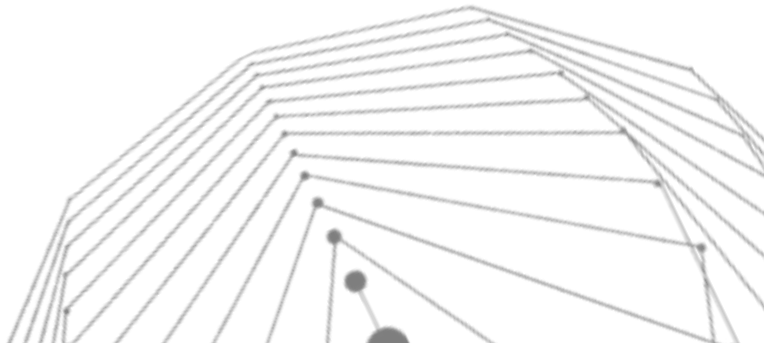
Eg.: *refer problem 8 in AIME 2018 II*, rather than in their original form.

By High school students.



The Irrationality proved.

The Problem: Prove that $\sqrt{2}$ cannot be expressed as a fraction of two integers; that is, prove it is irrational.

(Source: Internet)

This is truly a classic proof. It's one of the first times a result felt like pure magic to me. We are going to use a powerful technique called **Proof by Contradiction** (*reductio ad absurdum*). Instead of proving the statement directly, we'll assume the opposite is true and watch the logic completely fall apart.

Let's start with a bold assumption: let's pretend that $\sqrt{2}$ *can* be written as a fraction.

$$\sqrt{2} = \frac{p}{q}$$

where p and q are integers and $q \neq 0.$ We can also say that this fraction is in its **simplest form or sometimes reduced form**, meaning p and q have no common factors other than 1. This is a crucial detail.

Okay, let's play with this equation. Squaring both sides seems like a good next step to get rid of the square root:

$$2 = \frac{p^2}{q^2}$$

$$2q^2 = p^2$$ << look into this

What does this tell us? It says that p2 is equal to 2 times something, which means p2 must be an even number. Now, if the square of a number is even, the number itself must also be even. (Think about it: an odd number squared is always odd, like $3^2 = 9$)

So, we know for a fact that p is even. This means we can write p as 2k for some integer k. Let's substitute this back into our equation:

$$2q^2 = (2k)^2$$

$$2q^2 = 4k^2$$

Now, we can divide both sides by 2:

$$q^2 = 2k^2$$

But wait a second. This looks familiae. This equation is telling us that q2 is an even number, which must mean that q itself is also an even number.

Here's where our logic breaks down. We started by assuming that the fraction $\frac{p}{q}$ was in its simplest form. But we just proved that p is even and q is also even. If they are both even, they share a common factor of 2, which means the fraction wasn't simple at all!

This is a **contradiction.** Our initial assumption,, that $\sqrt{2}$ could be written as a fraction; has led us to an impossible conclusion. It's like building a beautiful house that collapses because the foundation was faulty. The faulty foundation here was our initial assumption.

Therefore, the assumption must be false. $\sqrt{2}$ cannot be expressed as a fraction. It is irrational.

The AM~GM inequality (but for two variables.)

$$AM \overset{\geq}{\underset{\leq}{?}} GM$$

The Problem: For any non-negative real numbers a and b, prove that their arithmetic mean is greater than or equal to their geometric mean.

$$\frac{a+b}{2} \geq \sqrt{ab}$$

(Source: This is a fundamental type inequality. The general form for n variables was proven by Augustin-Louis Cauchy.)

Solution: This inequality is one of the most useful tools in mathematics for comparing sums and products. The proof is surprisingly simple and elegant. It all starts from one simple, undeniable truth.

For any real number, its square is never negative.

$$x^2 \geq 0$$

This is a solid rock we can build our proof on. Let's be clever about what we choose for x. Since a and b are non-negative, their square roots are real numbers. Let's try setting $x = \sqrt{a} - \sqrt{b}$.

$$\boxed{(\sqrt{a} - \sqrt{b})^2 \geq 0}$$

Now, let's expand the left side:

$$(\sqrt{a})^2 - 2\sqrt{a}\sqrt{b} + (\sqrt{b})^2 \geq 0$$

$$a - 2\sqrt{ab} + b \geq 0$$

This is starting to look like what we want isn't it? Now, it's just a little bit of algebra to get to the finish line. Let's add $2\sqrt{ab}$ to both sides:

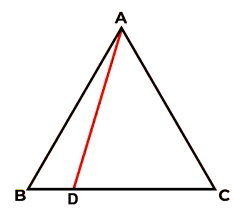
$$a + b \geq 2\sqrt{ab}$$

Finally, divide by 2: $\frac{a+b}{2} \geq \sqrt{ab}$

And there it is. We started from an unignorable fact and arrived directly at the result.

As a final thought, when do the equality hold? The '=' sign is only true if our starting "rock' was actually 0. This happens when $\sqrt{a} - \sqrt{b} = 0$ which means $\sqrt{a} = \sqrt{b}$, or simply a=b. So, the arithmetic mean equals the geometric mean only when the two numbers are the same.

The triangle inequality. (euclid's elements)



The Problem: Show that for any triangle, the sum of the lengths of any two sides must be greater than the length of the remaining side.

(Source: Euclid's *Elements*, Book 1, Proposition 20.)

Solution: This feels incredibly intuitive. If you want to walk between two points, the straight path is always the shortest. Any detour (the other two sides of a triangle) will make the journey longer. But intuition isn't proof, so let's make it rigorous.

Let's consider a triangle △ABC with side lengths a,b, and c opposite the corresponding vertices. We want to prove that a+b>c.

Let's use a geometric construction. Extend the side AC to a point D such that CD=CB=a. Now we've created a new, larger triangle △ABD.

Because we built it this way, △BCD is an isosceles triangle (since two sides, CB and CD, are equal to a). In an isosceles triangle, the angles opposite the equal sides are equal. So, ∠CBD = ∠CDB.

Now look at the big triangle, ∆ABD. The angle ∠ABD is clearly larger than ∠CBD, because it's made up of ∠ABC + ∠CBD. So we can say:

∠ABD ＞ ∠CBD

And since ∠CBD ＝ ∠CDB, we can substitude to get:

This means: ∠ABD ＞ ∠ CDB

A fundamental rule in geometry is that in any triangle, the longer side is the side in front of the largest Angle. In ∆ABD, the side opposite ∠ABD is AD. The side opposite ∠CDB (which is the same as ∠ADB) is AB.

Because ∠ABD ＞ ∠ADB, it must be true that the side opposite it is longer:

AD ＞ AB

What is AD? From our construction, we know AD=AC+CD. We also know AC=b and we set CD=a. So, AD=b+a. And of course, AB=c. Substituting these back in gives us:

B + a > c

And that completes the proof. The shortest distance between two points is a straight line. Let's get into another topic again. Yes!

The Pigeonhole Principle Basics

1. **In any group of 13 people, prove that at least two of them must have their birthday in the same month.**

(Source: This is a classic application of a combinatorial principle often attributed to Johann Peter Gustav Lejeune Dirichlet, but the accuracy may vary.)

Solution: This is a fantastic example of a powerful idea that sounds almost too simple to be useful. It's called the **Pigeonhole Principle.** In its most basic form, it says:

If you have more pigeons than pigeonholes, at least one pigeonhole must contain more than one pigeon.

It's unavoidable. If you have 5 pigeons and only 4 boxes to put them in, one box is going to get a crowd.

So, how do we apply this to our problem? We just need to identify the "pigeons" and the "pigeonholes".

** The **pigeons** are the 13 people.

** The **pigeonholes** are the 12 mo. of the year.

We are trying to place 13 pigeons into 12 pigeonholes. Since we have more people (pigeons) than months (Pigeonholes), the principle guarantees that at least one month must contain more than one person's birthday.

It's that simple. The proof isn't anything about calculation; it's about a logical understanding and clarification.. The principle doesn't tell us *which* month has the matching birthdays or how many people share it, only that a shared birthday month *must* exist somehow.

What if there were only 12 people? Would the principle still guarantee at least two shared birthdays? **If there were only 12 people, the Pigeonhole Principle wouldn't guarantee at least two shared birthdays. You could, in theory, have a scenario where each of the 12 people has their birthday in a different month, with no month repeated.Piossible, but not guaranteed.**

The sum of first *i* number of integers.

The Problem: Prove that for any positive integer i, the sum of all integers from 1 to i is given by the formula:

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